Linear First-order System

Consider a linear first-order system that is described by

$$z(t+1) = az(t) + bu(t), \quad t = 0, 1, \ldots, N$$

$z(t)$ is called the system state, and $u(t)$ is the system input. In physical world, $u(t)$ is fed to drive the system, where you can observe its state to evolve over time.

When $z(0) = z_0$, derive $z(t)$ as a function of z_0 and $u(s)$ for $s = 0, 1, t-1$.

By setting $z_0 = 0$, you will see that $z(t)$ is a linear combination of $u(s)$ for $s = 0, 1, \ldots, t-1$. Use this relationship to construct a linear system: $Au = z$ where $z = (z(1), z(2), \ldots, z(N))$ and $u = (u(0), u(1), \ldots, u(N-1))$.

From this result, when z is given (you plan to drive the system'state to a desired point), the problem is to find u that drive z to a given point.

Implementation

TODO: Fill in the following function:

```
def cal_input_firstordersys(z, a, b):
  return A, u
```

Constraints

Input parameters:

- $a \in (-1, 1)$
- $b \in (-5, 5)$
- $z \in \mathbf{R}^N$ where $N \geq 5$

Output answers:

- $A \in \mathbf{R}^{N \times N}$
- $u \in \mathbf{R}^N$